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Beliefs about the Stock Market and Investment Choices: Evidence from a Survey and a Field Experiment

Contents

1 Big picture

- **Question.** Do retail investors' beliefs about the autocorrelation of aggregate stock returns shape their return expectations and their real portfolio choices?
- **Setting.** The paper surveys clients of a German online bank and links survey beliefs to administrative data on account holdings and trades. This link is powerful because beliefs are elicited directly, while investment choices are measured from actual broker records rather than self-reports.
- **Belief object.** Investors report what they believe the average 12-month-ahead return of the German stock market has been after different 12-month past-return intervals. These beliefs reveal whether the investor sees stock returns as persistent, mean-reverting, or approximately unpredictable.
- **Main descriptive fact.** Investors disagree substantially about return dynamics. Many believe in mean reversion after high returns or extrapolation after low returns, even though historical German stock-market autocorrelation is close to zero.
- **Experimental intervention.** Half of respondents receive information showing that, historically, average 12-month-ahead returns are similar across past-return bins. This treatment targets beliefs about return autocorrelation.
- **Belief response.** Treated investors update beliefs in the direction implied by the historical information. The learning rate is partial: people do not move one-for-one toward the information, but the effect is large and statistically meaningful.
- **Investment-choice response.** Survey priors predict actual trading in historical account data. Investors who believe recent returns predict future returns trade in the direction implied by those beliefs.
- **Field-experimental outcome.** The treatment changes investors' equity purchases during the COVID-19 crash months after the survey, in the direction predicted by their updated return-extrapolation or mean-reversion beliefs.
- **Mechanism.** The intervention changes return expectations, and changed expectations affect trading. This provides causal evidence on a channel from belief formation to disagreement and trade.
- **Takeaway.** Subjective beliefs about return dynamics are not just survey noise. They are heterogeneous, manipulable, persistent enough to matter, and predictive of actual asset-market behavior.

2 Incremental contribution of this paper

- **From beliefs to actual trades.** A large literature documents subjective return expectations, but many studies rely on survey intentions or portfolio shares observed only coarsely. This paper directly links belief elicitation to administrative brokerage transactions, allowing the authors to study realized buying and selling decisions.
- **A specific belief dimension: perceived return autocorrelation.** Rather than asking only for an unconditional expected return, the paper measures how investors think returns vary with recent market performance. This isolates a central mechanism behind extrapolation, mean reversion, and disagreement.
- **Causal variation in belief structure.** The field experiment provides information about historical return predictability and shows that investors update their return expectations and later trading behavior. This goes beyond cross-sectional correlations between beliefs and portfolios.
- **Field experiment with delayed real outcome.** The intervention occurs before the COVID-19 crash, and the paper measures treatment effects on actual crash-period trades. This is a rare field setting in which a belief-treatment can be connected to an economically meaningful market episode months later.

- **Disagreement as a causal object.** The paper shows that heterogeneous beliefs about dynamics cause disagreement in expected returns. It therefore contributes to asset-pricing models in which disagreement and trading volume arise from heterogeneous interpretations of the same public return history.
- **Distinction between extrapolators and mean reverters.** The paper does not treat "retail investors" as a homogeneous behavioral group. Investors differ in whether they perceive continuation or reversal after past market movements, and these differences predict opposite trading responses.
- **Contribution to household finance.** The paper identifies a behavioral determinant of household trading that is measurable before the trading decision and can be experimentally shifted. This helps connect expectation-formation research to portfolio choice.

3 Data, survey design, and measurement

3.1 Survey population and administrative brokerage data

- The authors survey retail investors at a German online bank and link the survey responses to administrative account data.
- The administrative data contain holdings, transaction histories, and trading behavior. This allows the authors to observe equity purchases, sales, and net buying activity before and after the survey.
- The main survey sample contains investors who responded to the survey and for whom account-level data are available. Table 1 compares them with a benchmark household sample and checks treatment-control balance.
- The sample is investment-relevant: respondents are actual stock-market participants with brokerage accounts, equity holdings, and trading histories.

Interpretation: The key empirical advantage is that beliefs are measured at the individual level and can be matched to actual decisions. This avoids the common weakness of survey-only work, in which stated beliefs may not connect to real portfolio behavior.

3.2 Eliciting beliefs about stock-market autocorrelation

- Respondents see six bins for the DAX return over the previous 12 months: very negative, moderately negative, slightly negative, slightly positive, moderately positive, and very positive past returns.
- For each bin, respondents report their belief about the average DAX return over the subsequent 12 months.
- The pattern of answers across bins reveals whether a respondent perceives positive autocorrelation, no predictability, or mean reversion.
- A respondent who reports high future returns after high past returns and low future returns after low past returns is an extrapolator.
- A respondent who reports low future returns after high past returns and high future returns after low past returns is a mean-reverter.

Interpretation: This elicitation is richer than a single expected-return question. It measures the subjective mapping from recent market performance to future expected returns.

3.3 Gain-loss difference and belief classification

A central summary statistic is the perceived difference between expected future returns after positive versus negative past returns. A stylized version is:

$$Diff_i = E_i[R_{t+12} \mid R_t^{past} > 0] - E_i[R_{t+12} \mid R_t^{past} < 0]. \tag{1}$$

- $Diff_i > 0$ indicates extrapolative beliefs.
- $Diff_i < 0$ indicates mean-reversion beliefs.
- $Diff_i \approx 0$ indicates a belief that recent returns are not informative.
- The paper uses cutoffs such as $Diff_i \geq 4$ pp for extrapolators and $Diff_i < -4$ pp for mean reverters.

Interpretation: This statistic collapses a six-bin belief object into a tractable measure of perceived return autocorrelation.

3.4 Information treatment

- Treated respondents see the same dynamic figure as in the prior elicitation, but now the figure displays the actual historical average 12-month-ahead returns following each past-return interval.
- The historical evidence shows that the average subsequent return is close to flat across past-return bins.
- The control group does not receive this information at that point.
- The treatment therefore directly targets the perceived predictability of returns from recent market performance.

Interpretation: The intervention is tightly linked to the belief object. It does not provide general financial education; it provides targeted information about the low historical autocorrelation of stock returns.

3.5 Outcome variables in administrative data

The main trading variable is relative net buying volume:

$$Relative\ net\ buying\ volume_{i,t} = \frac{Amount\ purchased_{i,t} - Amount\ sold_{i,t}}{Financial\ wealth_{i,t-1}}. \tag{2}$$

Other outcomes include:

- probability of being a net buyer;
- relative gross buying volume;
- relative gross selling volume;
- probability of trading;
- relative net buying conditional on trading.

Interpretation: Relative net buying volume approximates the active change in equity exposure due to trading, abstracting from passive changes caused by price movements.

4 Empirical specifications

4.1 Perception gap and expectation updating

The perception gap is the difference between historical information and the respondent's corresponding prior:

$$PerceptionGap_i = ActualHistoricalReturn_{bin(i)} - PriorHistoricalReturn_{i,bin(i)}. \tag{3}$$

The main updating regression is:

$$Updating_i = \alpha_0 + \alpha_1 PerceptionGap_i \times Treatment_i + \alpha_2 PerceptionGap_i + \alpha_3 Treatment_i + \Gamma X_i + \varepsilon_i. \tag{4}$$

- $Updating_i$ is the difference between posterior and prior expected 12-month-ahead return.
- α_1 is the learning rate: it captures how strongly treated investors move toward the information.
- Some specifications instrument the perception gap based on perceived past returns with a version based on the actual realized past return.

Interpretation: If the information truly changes beliefs, treated respondents with larger perception gaps should update more. This is precisely what the interaction term measures.

4.2 Correlational trading specification

The paper predicts each investor's expected return using his perceived historical autocorrelation and the realized past return at each month. It then estimates:

$$Y_{i,t} = \alpha_0 + \alpha_1 \Delta E_{i,t}(hist.R_{12m} \mid previousR_{12m}) + \alpha_2 E_{i,t-1}(hist.R_{12m} \mid previousR_{12m}) + \Gamma X_{i,t-1} + \mu_t + \varepsilon_{i,t}. \tag{5}$$

- $Y_{i,t}$ is a trading outcome such as relative net buying volume.
- $E_{i,t-1}$ is the predicted return expectation at the end of the previous month.
- $\Delta E_{i,t}$ is the change in the prediction over the current month.
- μ_t are month-year fixed effects.

Interpretation: This specification asks whether the time variation in predicted return expectations implied by an investor's own belief model predicts the investor's actual trading.

4.3 Crash-period treatment-effect specification

For the COVID-19 crash, the paper estimates event-time difference-in-differences specifications of the form:

$$\begin{aligned} Y_{i,t} = \beta_0 + \beta_1 Crash_t \times Treatment_i + \beta_2 PostsurveyPrecrash_t \times Treatment_i \quad (6) \\ + \text{individual fixed effects} + \text{time fixed effects} + \Gamma X_{i,t-1} + \varepsilon_{i,t}. \quad (7) \end{aligned}$$

- The crash period is the weeks around the sharp COVID-19 stock market decline.
- The paper separates investors whose priors imply they should revise expected returns downward during the crash from those whose priors imply upward revisions.
- The key test is whether the treatment shifts crash-period buying in opposite directions for these two groups.

Interpretation: This is the field-experimental test of the belief channel. The treatment occurred before the crash, and the outcome is later real trading behavior.

5 Identification and empirical design

5.1 Randomized information treatment

- Treatment is randomly assigned among survey respondents.
- Table 1 shows balance across demographic characteristics, portfolio characteristics, financial literacy, and pretreatment beliefs.
- The treatment varies only the information about historical return predictability, which supports a causal interpretation of belief updating.

Interpretation: Randomization identifies the causal effect of providing historical return-predictability information on posterior beliefs.

5.2 Linking beliefs to trades

- Administrative brokerage records remove concerns that respondents exaggerate or misreport their trading.
- Pre-survey trading behavior provides correlational evidence that belief heterogeneity matters for actual choices.
- Post-treatment crash-period trading provides causal evidence that a belief intervention affects later investment choices.

Interpretation: The paper combines survey evidence, experimental variation, and administrative data. This triangulation is the main source of credibility.

5.3 Why the COVID-19 crash is useful

- The crash generated a large and salient negative return shock after the survey but before the belief treatment had become irrelevant.
- Extrapolators and mean reverters should interpret the same crash differently.
- Treated investors should have weaker extrapolative or mean-reverting reactions because the treatment teaches that past returns have little predictive power.

Interpretation: The crash creates a natural stress test of the belief-treatment mechanism.

5.4 Threats and limitations

- Respondents are clients of an online bank, so the sample is more financially active than the general population.
- Survey participation is voluntary, so respondents may differ from nonrespondents.
- The treatment could affect salience or confidence, not only perceived historical autocorrelation.
- The COVID-19 crash was an unusual event, so treatment effects may differ in normal periods.
- The treatment shifts beliefs partially, so estimates are reduced-form effects of a realistic information intervention rather than full-information effects.

Interpretation: These limitations matter, but the paper's strength is that the causal chain is observed: treatment changes beliefs, and those belief changes predict real trading in the expected direction.

6 Tables and figures

6.1 Figure 1 and Table 1: experimental interface and sample balance

Read:

- Figure 1 shows the dynamic belief-elicitation interface and the information-treatment screen.
- Respondents report their own beliefs first and, if treated, then see the historical average returns for each past-return bin.
- Table 1 shows that the treatment and control groups are balanced on demographic, portfolio, financial-literacy, risk-tolerance, and belief variables.

Economic message: The experiment targets a clearly measured belief object, and randomization appears balanced across observable characteristics.

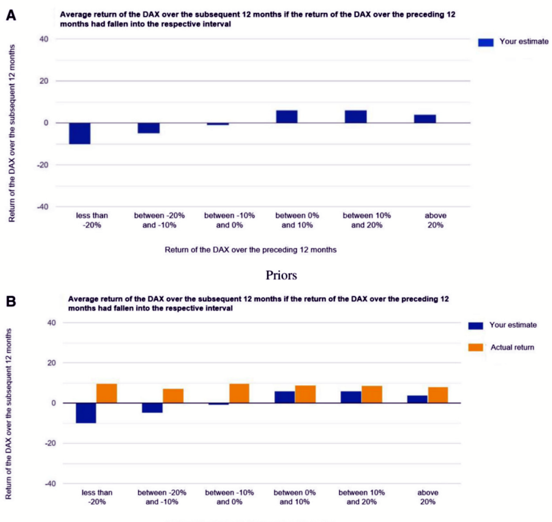


Figure 1: survey elicitation and treatment screens

Table 1
Summary statistics and balance check

	PHF	Online brokerage sample							
	(1) 2017 Mean	(2) Mean	(3) Median	(4) SD	(5) p25	(6) p75	(7) Treatment group: Mean	(8) Control group: Mean	(9) p-valu (7) = (
Female	0.49	0.16	0.00	0.37	0.00	0.00	0.17	0.16	.511
Age	50.55	45.24	45.00	14.15	34.00	55.00	45.84	44.66	.067
University	0.36	0.54	1.00	0.50	0.00	1.00	0.52	0.56	.079
Employed	0.65	0.77	1.00	0.42	1.00	1.00	0.75	0.78	.150
Household net income	3,808	3,914	4,000	2,769	2,000	5,250	3,927	3,902	.837
Household net wealth	361,783	300,488	125,000	458,044	12,500	375,000	307,809	293,294	.483
Total financial wealth at bank		55,272	22,082	98,312	5,581	65,752	55,073	55,468	.929
Portfolio value at bank		43,970	14,872	87,671	3,726	47,620	43,438	44,489	.795
Equity holdings at bank		39,405	13,381	78,678	3,437	42,458	38,318	40,467	.553
Average monthly equity trades		1.73	0.67	3.29	0.00	2.00	1.75	1.71	.812
Risk tolerance (1-7)		4.56	5.00	1.17	4.00	5.00	4.54	4.58	.403
Trading experience (years)		14.13	15.00	10.87	4.00	20.00	14.38	13.88	.309
Financial literacy score (0-3)		1.82	2.00	0.78	1.00	2.00	1.81	1.83	.543
Follow DAX developments (1-7)		4.76	5.00	1.81	3.00	6.00	4.78	4.75	.737
Investment horizon $\geq$ 5 years		0.49	0.00	0.50	0.00	1.00	0.48	0.50	.373
Perceived return last 12 months		5.09	5.00	6.07	2.00	8.00	4.99	5.19	.475
Confident in perceived return		0.64	1.00	0.48	0.00	1.00	0.64	0.64	.855
Expected return next 12 months		3.21	4.00	6.28	1.50	6.00	3.32	3.09	.423
Confident in expected return		0.54	1.00	0.50	0.00	1.00	0.54	0.53	.494
Perceived mean hist. ret. intervals		8.36	7.83	4.62	5.17	11.17	8.39	8.32	.739
Perceived diff. gain-loss historical		–4.88	–4.67	9.64	–10.67	0.50	–4.68	–5.08	.360
Extrapolator (diff. $\geq$ 4)		0.16	0.00	0.36	0.00	0.00	0.16	0.15	.725
Mean-reverter (diff. < -4)		0.53	1.00	0.50	0.00	1.00	0.53	0.52	.789
In follow-up sample		0.46	0.00	0.50	0.00	1.00	0.46	0.46	.886
p-value ((7) = (8) for all variables)									.745
Observations		1,961					972	989	

This table shows summary statistics for the sample of retail investors at the online bank that responded to our main survey (columns 2–6), as well as benchmarks from the German population of individuals participating in the stock market as measured in the 2017 wave of the Bundesbank's Panel of Household Finance (column 1). Columns 7–9 provide a check of balance of means between treatment and control group, showing means of

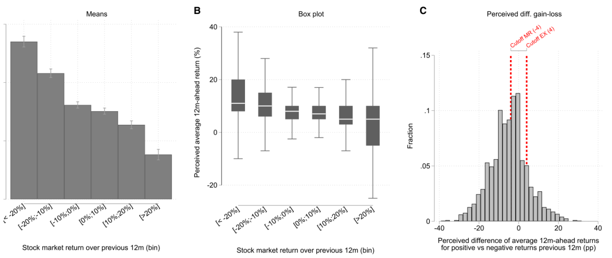
Table 1: summary statistics and balance check

6.2 Figure 2 and Table 2: prior beliefs and manipulation check

Read:

- Figure 2 documents large heterogeneity in beliefs about the historical autocorrelation of stock returns.
- Many respondents believe in mean reversion, while others believe in extrapolation or no predictability.
- Table 2 shows that the treatment increases agreement with statements that recent returns do not predict future returns.
- Treatment effects differ by prior type, especially for mean reverters and extrapolators.

Economic message: Investors disagree about a basic statistical property of returns, and targeted information moves their beliefs in the intended direction.



efs about the autocorrelation of aggregate returns

e summarizes prior beliefs about the autocorrelation of returns of the German stock market in the last 50 years among respondents to our main survey. Panel A shows the sample espondents' beliefs about average 12-month-ahead stock returns for six intervals of realized returns over the previous 12 months. Panel B displays box plots of respondents' prior out average 12-month-ahead stock returns for the six intervals of realized returns over the previous 12 months, including the median, 25th percentile, and 75th percentile for each anel C shows a histogram of respondents' perceived difference in average 12-month-ahead returns between the positive and the negative realized return scenarios, including the use to define mean reverters, neutrals and extrapolators.

Figure 2: prior beliefs about autocorrelation

Table 2
Manipulation check

	Positive return irrespective of previous return (z)		No sense to buy after high return (z)		Positive return more likely after high return (z)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.092**		–0.054		–0.147***	
	(0.044)		(0.044)		(0.045)	
Treatment × Extrapolator (diff. ≥ 4) (a)		–0.037		0.021		–0.375***
		(0.112)		(0.114)		(0.115)
Treatment × Neutral (–4 ≤ diff. < 4)		0.078		0.075		–0.084
		(0.079)		(0.080)		(0.081)
Treatment × Mean-reverter (diff. < –4) (b)		0.140**		–0.155***		–0.114*
		(0.059)		(0.060)		(0.062)
Extrapolator (diff. ≥ 4)	–0.035	0.022	–0.018	0.008	0.143**	0.288***
	(0.069)	(0.097)	(0.071)	(0.098)	(0.072)	(0.102)
Mean-reverter (diff. < –4)	–0.066	–0.097	0.046	0.160**	–0.127**	–0.113
	(0.050)	(0.071)	(0.051)	(0.070)	(0.053)	(0.072)
p-value (a=b)		.161		.174		.047
Observations	1,961	1,961	1,961	1,961	1,961	1,961
R-squared	.10	.10	.08	.08	.04	.04

This table shows estimations of the effect of the information treatment on posterior agreement with verbal statements describing beliefs about the autocorrelation of aggregate returns among respondents to our main survey. Agreement with the statements is elicited on seven-point categorical scales and is z-scored using the means and standard deviations in the sample. The statements are: "With an investment in stocks one can expect a positive return, independently of how the stock market has developed in the recent past" (columns 1 and 2); "When the stock market has recently increased it makes no sense to buy stocks" (columns 3 and 4); "When the

Table 2: manipulation check

6.3 Figure 3 and Table 3: expectation updating

Read:

- Figure 3 shows that predicted returns implied by perceived autocorrelation are positively related to investors' reported expected returns.
- Table 3 estimates learning rates around 0.09–0.14, depending on whether updating is measured using point beliefs or distribution means and whether IV is used.
- Treated respondents update their expected returns in proportion to their perception gap.

Economic message: Beliefs about historical predictability matter for expected returns, and the information treatment causally shifts return expectations toward the historical evidence.

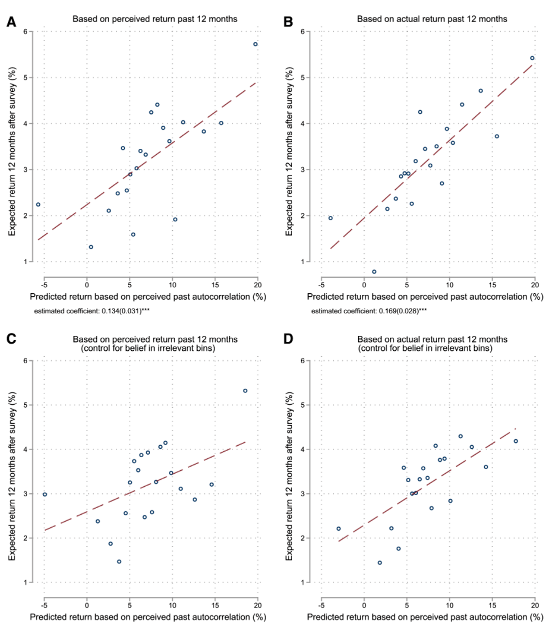


Figure 3: expected returns versus predicted expectations

Table 3
Updating of 12-month-ahead return expectations

	Updating (point belief)		Updating (mean distr.)	
	(1) OLS	(2) IV	(3) OLS	(4) IV
Treatment × Perception gap	0.086**	0.138***	0.115***	0.142**
	(0.038)	(0.051)	(0.044)	(0.060)
Perception gap	–0.004	–0.019	0.022	0.044
	(0.025)	(0.033)	(0.028)	(0.038)
Treatment	1.077***	1.007***	0.047	0.019
	(0.212)	(0.219)	(0.263)	(0.266)
First-stage F-stat		1,020.48		1,020.48
Observations	1,961	1,961	1,961	1,961
R-squared	.05	.04	.04	0.04

This table examines changes in expectations about aggregate stock returns over the 12 months after the survey in response to the information among respondents to our main survey based on estimations of specification 2. The outcomes are the difference between posterior and prior point expectations about the 12-month-ahead return (columns 1 and 2) and the difference between the mean of the respondent-level posterior distribution over 12-month-ahead returns and the prior point expectation (columns 3 and 4). The perception gap is based on the

Table 3: updating of 12-month-ahead return expectations

6.4 Figure 4 and Table 4: disagreement in return expectations

Read:

- Figure 4 shows prior and posterior distributions by treatment arm.
- The treatment compresses the distribution of posterior expected returns.
- Table 4 shows that the p90–p10 range stays at 15 pp in the control group but falls to 10.60 pp in the treatment group.
- The cross-sectional standard deviation of posteriors is also significantly lower in the treatment group.

Economic message: Heterogeneous beliefs about return dynamics are a causal driver of disagreement in expected returns.

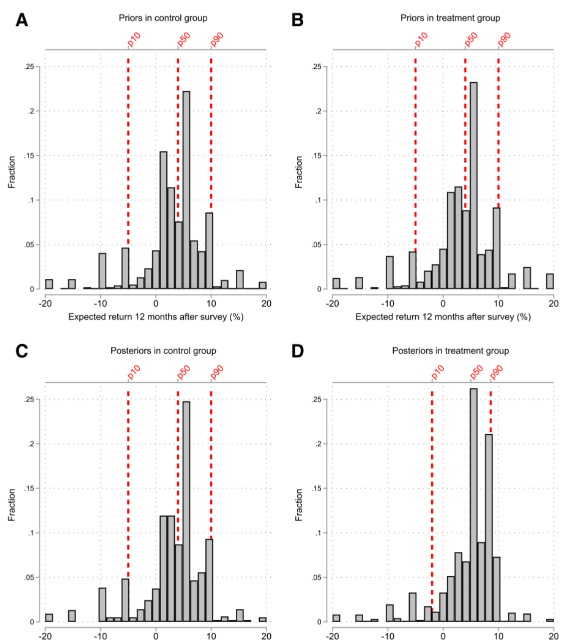


Figure 4: priors and posteriors

Table 4
Disagreement in priors and posteriors

	Expected return next 12 months		
	(1) Control group	(2) Treatment group	(3) *p*-value (1) = (2)
p90 − p10 priors	15.00	15.00	.000
p90 − p10 posteriors	15.00	10.60	.000
SD priors	6.10	6.45	.241
SD posteriors	5.86	5.29	.008
Observations	989	972	

This table displays the difference between the 90th and the 10th percentiles and the cross-sectional standard

Table 4: disagreement in priors and posteriors

6.5 Table 5: beliefs and pre-survey trading behavior

Read:

- The perceived conditional historical return positively predicts the probability of net buying and relative net buying volume.
- In the full sample, a higher predicted return is associated with greater net purchases of equity.
- The relationship is stronger in the active trading sample.
- Results hold with month fixed effects and controls for portfolio characteristics and demographics.

Economic message: Investors' subjective models of return dynamics explain actual pre-treatment trading behavior.

Table 5
Perceived autocorrelation and trading behavior

	Probability net buy (%)	Relative net buying volume (%)	Relative gross buying volume (%)	Relative gross selling volume (%)	Probability trade (%)	Probability net buy (%)	Relative net buying volume (%
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Full sample							
Δ Perceived conditional hist. return (%)	0.050* (0.029)	0.015** (0.007)	0.011** (0.006)	–0.005 (0.005)		0.096 (0.062)	0.042* (0.024)
Lagged perceived conditional hist. return (%)	0.025 (0.042)	0.010* (0.006)	0.010 (0.007)	0.001 (0.007)	0.006 (0.043)	0.049 (0.068)	0.030 (0.020)
|Δ Perceived conditional hist. return (%)|					0.011 (0.079)		
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Specification						Cond. on trading	Cond. on trading
Observations	74,569	74,569	74,569	74,569	74,569	22,131	22,131
Number of investors	1,871	1,871	1,871	1,871	1,871	1,782	1,782
R-squared	.14	.03	.07	.04	.15	.08	.06
B. Active sample							
Δ Perceived conditional hist. return (%)	0.097** (0.044)	0.025*** (0.009)	0.022*** (0.007)	–0.003 (0.007)		0.113* (0.063)	0.048** (0.023)
Lagged perceived conditional hist. return (%)	0.044 (0.062)	0.016* (0.008)	0.022** (0.011)	0.008 (0.011)	0.027 (0.059)	0.033 (0.075)	0.036* (0.019)
|Δ Perceived conditional hist. return (%)|					0.042 (0.116)		
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Specification						Cond. on trading	Cond. on trading
Observations	46,056	46,056	46,056	46,056	46,056	18,855	18,855
Number of investors	1,841	1,841	1,841	1,841	1,841	1,639	1,639
R-squared	.11	.03	.08	.05	.11	.08	.05

This table examines the association between the perceived autocorrelation of returns and equity trading decisions among respondents to our main survey based on investor-month level estimations. The "relative net buying volume" is defined as the percent ratio of net purchases of equity over a given month relative to end-of-previous-

6.6 Table 6 and Table 7: treatment effects on crash-period trading

Read:

- Table 6 splits investors by whether their priors imply downward or upward expectation adjustments during the COVID-19 crash.
- The treatment shifts buying in opposite directions across these groups, as predicted by the belief mechanism.
- The difference in treatment effects is statistically significant for relative net buying and buying probability.
- Table 7 shows additional specifications for the full sample, including post-survey and crash-period interactions.

Economic message: The field experiment links beliefs to real portfolio decisions: changing beliefs about return autocorrelation changes how investors trade during a major market crash.

Table 6
Treatment effects on trading responses to the COVID-19 crash: Full sample

	Probability net buy (%)	Relative net buying volume (%)	Relative gross buying volume (%)	Relative gross selling volume (%)	Probability trade (%)	Probability net buy (%)	Relative net buying volume (%)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Predicted exp. adjustment ≤ 0							
Crash ×	2.480	0.290**	0.234*	0.004	1.642	7.849**	1.542**
Treatment (a)	(2.052)	(0.120)	(0.136)	(0.126)	(2.562)	(3.827)	(0.603)
Observations	15,639	15,639	15,639	15,639	15,639	1,866	1,866
Number of investors	411	411	411	411	411	281	281
R-squared	0.15	0.07	0.13	0.10	0.16	0.27	0.20
B. Predicted exp. adjustment > 0							
Crash ×	–2.374***	–0.063	–0.094	–0.028	–2.688**	–2.530	–0.312
Treatment (b)	(0.832)	(0.080)	(0.080)	(0.044)	(1.206)	(2.886)	(0.435)
Observations	55,584	55,584	55,584	55,584	55,584	6,746	6,746
Number of investors	1,432	1,432	1,432	1,432	1,432	1,007	1,007
R-squared	0.16	0.07	0.14	0.07	0.16	0.27	0.22
C. Difference in treatment effects							
(a) - (b)	4.861**	0.354**	0.329*	0.032	4.332	11.502**	1.947***
	(2.318)	(0.144)	(0.177)	(0.139)	(3.041)	(4.378)	(0.608)
D. Control group							
Crash ×	2.638	0.124	0.153	0.073	2.935*	5.219	1.076**
𝟙 (Predicted exp. adjustment > 0)	(1.787)	(0.113)	(0.117)	(0.081)	(1.737)	(3.711)	(0.462)
Observations	35,728	35,728	35,728	35,728	35,728	4,528	4,528
Number of investors	927	927	927	927	927	669	669
R-squared	0.15	0.07	0.14	0.07	0.16	0.23	0.20
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Specification						Cond. on trading	Cond. on trading

This table analyzes trading responses to the COVID-19 crash among respondents to our main survey on investor-week-level data. Panels A and B estimate treatment effects of our intervention on trading responses based on specification 5. Panel A focuses on respondents who would weakly downward adjust their return expectations in response to the crash if return expectations were exclusively based on the investor's prior perceived historical autocorrelation of returns, assuming accurate beliefs about the change in the realized return over the crash period. Panel B focuses on those who would upward adjust their return expectations in response to the crash based on their priors. Panel C displays differences in treatment effects on trading responses to the crash between these two groups. Panel D estimates control group respondents' trading adjustments to the crash depending on their

Table 6: treatment effects during the COVID-19 crash

Table 7
Additional results on treatment effects on trading behavior

	Relative net buying volume (%)				Probability trade (%)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post-survey × Treatment	–0.030				–0.405			
	(0.032)				(0.481)			
Crash × Treatment		0.014	0.066	–0.115		–1.731*	–0.652	–2.684**
		(0.068)	(0.117)	(0.150)		(0.994)	(1.205)	(1.313)
Post-survey precrash ×		–0.037	–0.069	0.039		–0.171	–0.588	0.075
Treatment		(0.031)	(0.042)	(0.053)		(0.462)	(0.632)	(0.645)
	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Full	Full	Pred. exp.	Pred. exp.	Full	Full	Abs. pred.	Abs. pred.
Sample			adj. pre-crash ≤ 0	adj. pre-crash > 0			exp. adj. crash ≤ Median	exp. adj. crash > Median
Observations	71,223	71,223	50,885	19,354	71,223	71,223	35,392	34,847
Number of investors	1,843	1,843	1,307	507	1,843	1,843	915	899
R-squared	.07	.07	.06	.08	.16	.16	.16	.16

This table presents additional results on treatment effects of our intervention on trading behavior. All estimations are based on investor-week-level data sets. The "relative net buying volume" is defined as the percent ratio of net purchases of equity over a given week relative to end-of-previous-week financial wealth. "Probability trade" is a dummy variable, multiplied by 100, indicating whether a respondent has a relative net buying volume of at least 1 pp in absolute terms. "Postsurvey" takes value one for the weeks from September 16, 2019, until March 13, 2020, and zero otherwise. "Crash" takes value one for the weeks from February 17, until March 13, 2020. "Postsurvey precrash" takes value one for the weeks from September 16, 2019, until February 16, 2020. Columns 1, 2, 5, and 6 focus on the full data set. Columns 3 and 4 focus on respondents who respectively would (weakly) downward adjust or upward adjust their return expectations over the postsurvey precrash period if return expectations were exclusively based on the investor's prior perceived historical autocorrelation of returns, assuming accurate beliefs

Table 7: additional treatment-effect results

6.7 Table 8: sources of beliefs

Read:

- Table 8 studies why some investors believe in mean reversion.
- Knowledge of research does not strongly explain mean-reversion beliefs.
- Belief in mean reversion is related to thinking of specific reversal episodes.
- The result supports the idea that memory and salient episodes shape subjective beliefs about return dynamics.

Economic message: Investors' beliefs about return predictability are not just abstract statistical beliefs; they are connected to recalled market episodes and narratives.

Table 8
Sources of beliefs: Memory and return predictability by valuation ratios

	Mean-reverter (diff. < -4)					
	(1)	(2)	(3)	(4)	(5)	(6)
Knowledge of research contributed to estimates	3.118					4.295
	(13.834)					(14.404)
Other financial variables contributed to estimates		–18.712*				–18.991*
		(10.619)				(10.489)
High knowledge of financial market theories			–0.158			–0.108
			(9.200)			(9.830)
Thought of specific reversal episode				20.393***		20.445***
				(6.759)		(6.964)
Thought of specific nonreversal episode					7.901	1.489
					(8.539)	(8.595)
Observations	208	208	208	208	208	208
R-squared	.13	.15	.13	.17	.14	.19

This table examines memory and beliefs about return predictability by valuation ratios as two potential sources of beliefs in mean reversion among German retail investors participating in an additional descriptive survey. The outcome is a dummy variable, multiplied by 100, indicating whether the respondent is classified as a mean reverter (perceived difference in average 12-month-ahead returns between the positive and the negative realized return scenarios lower than -4 pp). The main independent variables are the following: dummy variables indicating whether the respondent reports that knowledge of results from finance research or memory of past developments

7 Short summary

- Retail investors disagree sharply about whether stock returns extrapolate, mean-revert, or are unpredictable.
- These beliefs predict expected returns and actual trading choices.
- An information treatment about historical return autocorrelation shifts investors' beliefs and expected returns.
- The treatment reduces disagreement in posterior return expectations.
- Before the experiment, investors' perceived return predictability is correlated with trading in the administrative data.
- During the COVID-19 crash, the treatment changes actual equity purchases in the direction implied by the belief intervention.
- The paper provides causal evidence that beliefs about return dynamics drive disagreement and trade in asset markets.
- Memory of salient reversal episodes helps explain why some investors believe in mean reversion.

8 Conclusion

8.1 Core conclusion

Laudenbach, Weber, Weber, and Wohlfart show that retail investors' beliefs about return autocorrelation are heterogeneous, causally malleable, and economically relevant. These beliefs shape return expectations and actual equity trading behavior.

8.2 Economic intuition

Investors do not merely disagree about the unconditional level of future stock returns. They disagree about the law of motion of returns. A market crash therefore means different things to different investors: to an extrapolator, it predicts further losses; to a mean reverter, it predicts a rebound. By correcting beliefs about historical predictability, the experiment changes how investors interpret later market movements.

8.3 Takeaway

The paper brings together survey expectations, randomized information provision, and administrative trading data. Its central lesson is that subjective models of return dynamics are a concrete microfoundation for disagreement and trade in financial markets.